

Stage 1 Report

Flow Matching as a Layer

University of Haifa

Yousef Shihade, Mira Bitar

This report walks through the five Stage 1 notebooks in order -- what each one did, the graphs that best show we understood the data and the results, and why specific choices were made along the way. Together they answer every item in the assignment's Evaluation and Results section: the accuracy table, accuracy vs. training-set size, training curves, confusion matrices, and feature visualizations.

Protocol summary

Datasets	DTD (47 texture classes, official partition 1) and FGVC-Aircraft (100 variant-level classes)
Frozen encoders	ResNet-18 (512-d, both datasets); DINOv2 ViT-S/14 (384-d, Aircraft only)
Baselines	A trained linear probe ($s = Wz + b$), and image-derived class prototypes (no training)
Training sizes	K in {5, 10, full}. 5-/10-shot: 3 balanced random subsets. Full: complete training split, 3 seeds
Metric	Top-1 accuracy on the complete, untouched official test split, every time

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- Notebook 1 -- Data Setup & Protocol Verification
- Notebook 2 -- Feature Extraction
- Notebook 3 -- Linear Probe (required baseline; training curves)
- Notebook 4 -- Image-Derived Prototypes (second baseline)
- Notebook 5 -- Analysis (accuracy table, accuracy vs. K, confusion matrices, feature visualizations)

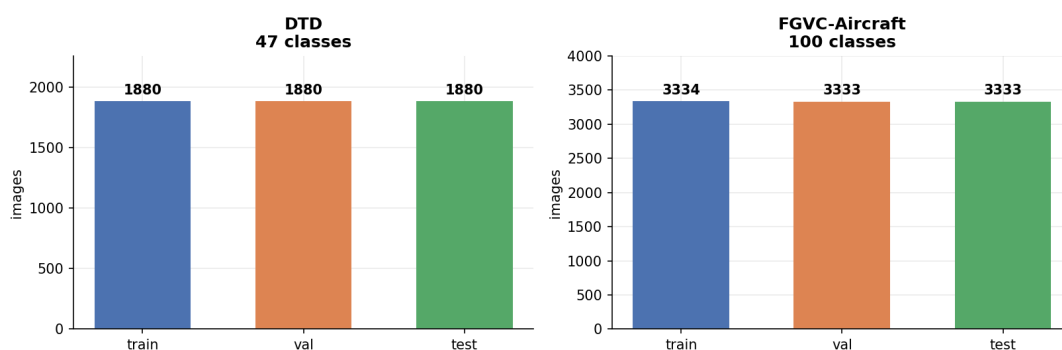
Notebook 1 — Data Setup & Protocol Verification

Loaded the official data splits and verified the protocol before extracting a single feature.

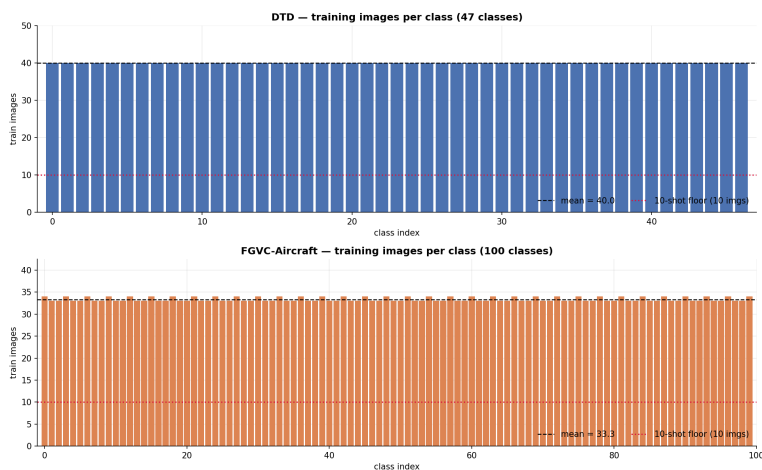
20 automated checks (10 per dataset) confirm the right partitions, split sizes, pairwise-disjoint splits, and file presence on disk -- all 20 pass before Notebook 2 ever runs an encoder.

Check category	DTD	FGVC-Aircraft
Class count & partition/level correct	PASS	PASS
Split sizes match official counts	PASS	PASS
Train/val/test pairwise disjoint	PASS	PASS
Every referenced file present on disk	PASS	PASS
>=10 training images per class (10-shot floor)	PASS	PASS

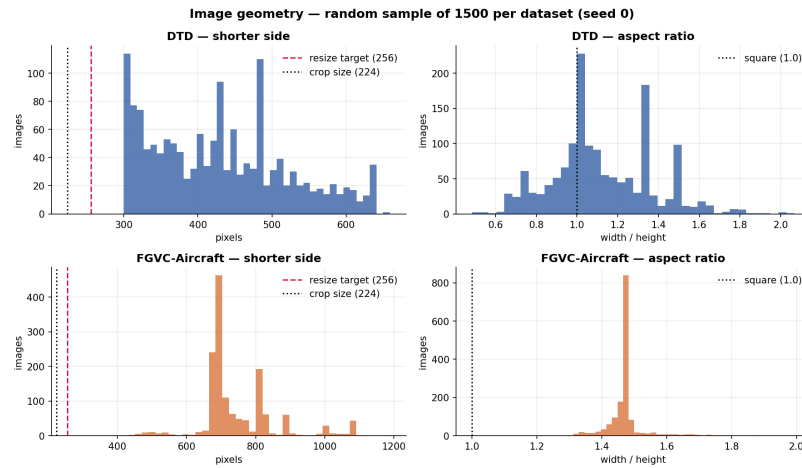
Official train / val / test splits — kept separate, never merged



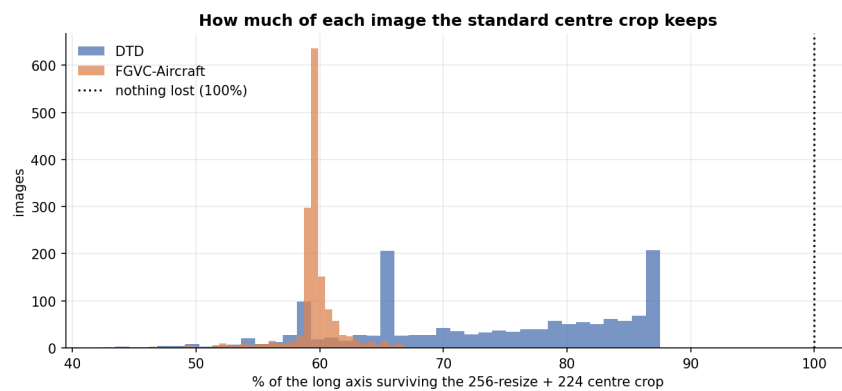
Official train/val/test sizes per dataset. Both datasets happen to split into near-equal thirds (~33% each) -- that proportion comes entirely from the dataset source, not a choice made here.



Training images per class. DTD is a flat 40/class; Aircraft averages ~33. Both sit comfortably above the red 10-shot floor for every single class.



Shorter-side and aspect-ratio distributions. Aircraft photos spike hard at aspect ratio ~ 1.47 (wide landscape shape) -- this specific shape is what makes the centre crop expensive for this dataset (next figure).

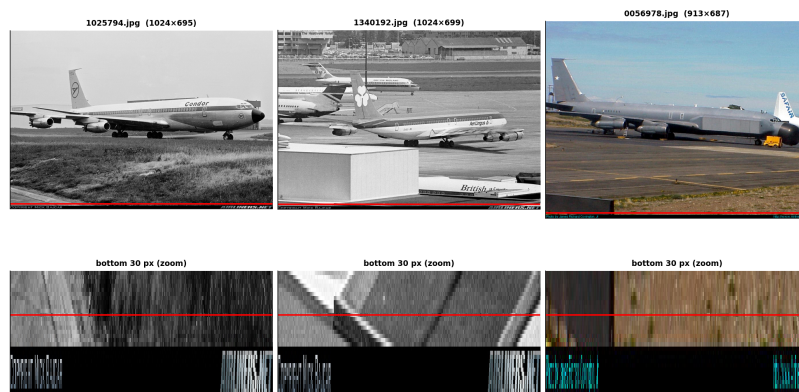


The standard 256-resize + 224 centre-crop keeps only $\sim 59\%$ of each aircraft image's width; DTD loses much less on average since textures are orientation-uniform.

Why this choice:

The pretrained checkpoint's own preprocessing (resize 256, centre-crop 224) is used unchanged for both datasets, even knowing it discards $\sim 41\%$ of each Aircraft image's width. Deviating from the checkpoint's published preprocessing would mean the encoder sees a different input distribution than it was pretrained on, breaking comparability with the frozen encoder itself and with Stages 2-3. This is recorded as a known, quantified limitation -- not something we changed.

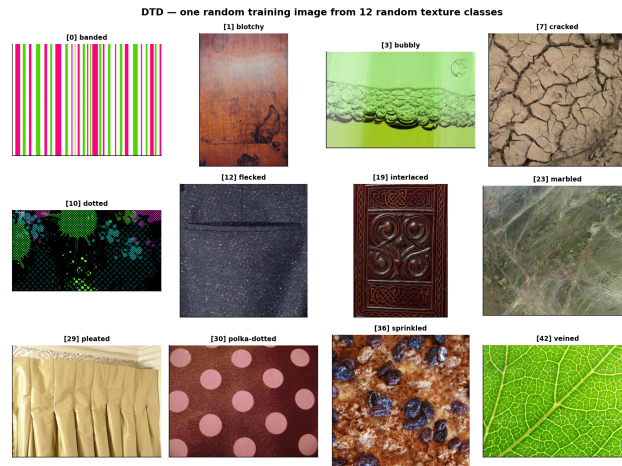
FGVC-Aircraft — burned-in copyright banner along the bottom 20 px (red line = recommended crop)



Every FGVC-Aircraft image carries a 20-pixel copyright banner (e.g. 'COPYRIGHT MICK BAJCAR') burned directly into the bottom-edge pixels.

Why this choice:

This banner is cropped before encoding (done in Notebook 2) -- and this is a different decision from the centre crop above, worth keeping separate. The banner is not part of the checkpoint's preprocessing; it is a printed artifact on the source images themselves, unrelated to what the pretrained model expects. Removing it is free, strictly more correct (the encoder sees a photo, not a photo plus a copyright notice), and does not touch comparability with the checkpoint. Notebook 2 then tests directly whether this crop actually changes accuracy (it does not, within noise) -- so it is kept as hygiene, not claimed as an improvement.



Visual sanity check: one random training image per class. Cheap but important -- catches a corrupted extraction or a label/image misalignment that accuracy numbers alone would hide.



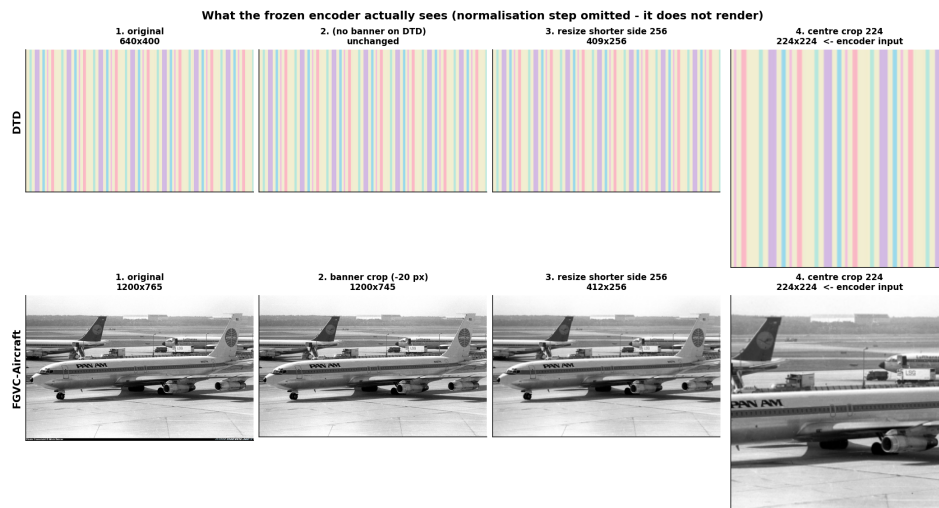
Training-set size at each K setting. For DTD the full set is 8x the 5-shot set; for Aircraft it's 6.7x -- enough spread that the three K settings stay clearly distinct.

Notebook 2 — Feature Extraction

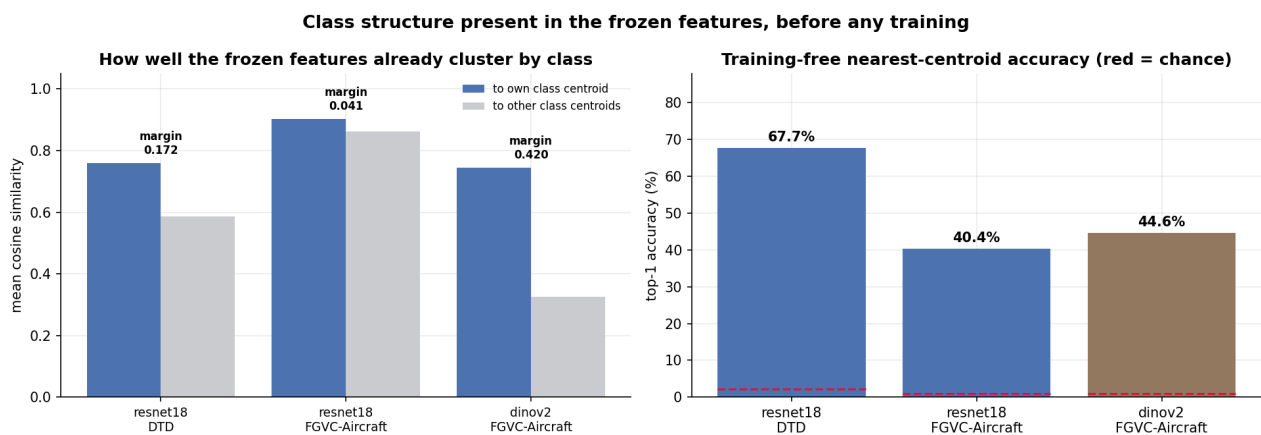
Ran both frozen encoders once over every image and cached the results -- nothing touches an image again after this.

9 cache files, 25,640 images encoded, all verified against the source labels after being reloaded from disk. Before any classifier was trained, the cached features alone were used to run diagnostics that predicted how the rest of the project would turn out.

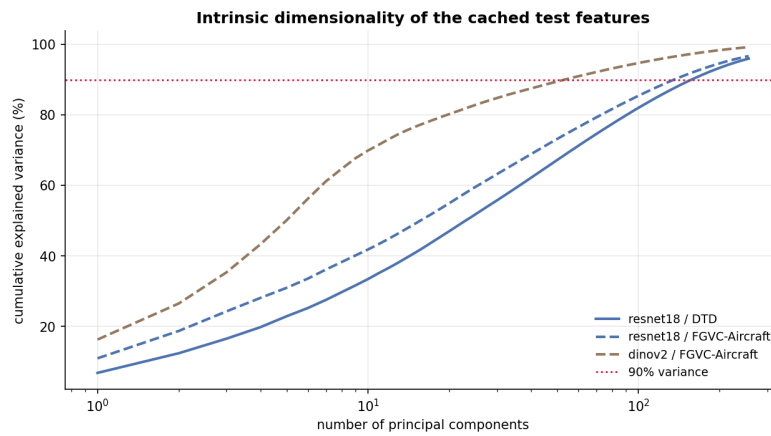
Encoder	Checkpoint	Feature dim	Params	Trainable
ResNet-18	torchvision IMAGENET1K_V1	512	11.18M	0
DINOv2 ViT-S/14	facebookresearch/dinov2	384	22.06M	0



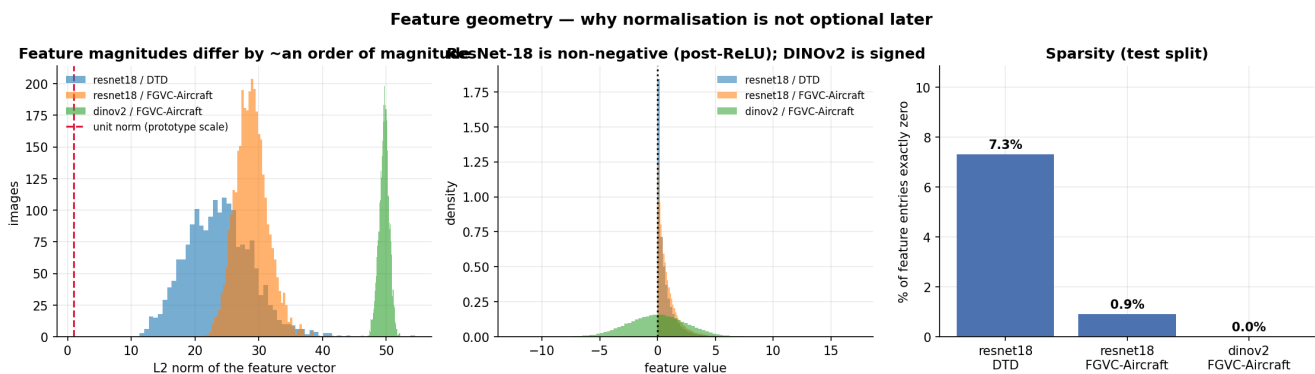
One real image of each dataset traced through every stage: original -> banner crop (Aircraft only) -> resize 256 -> centre-crop 224. This is exactly what the encoder receives -- shown, not just described.



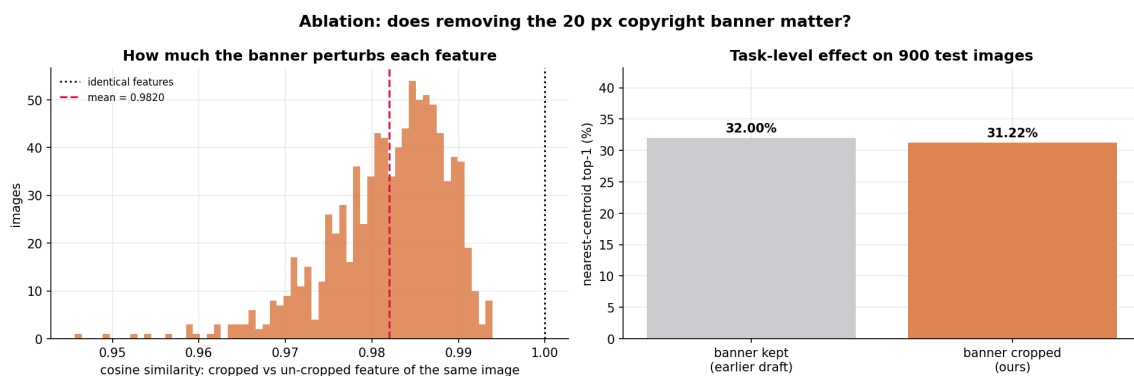
Cosine-similarity margin to own vs. other class centroids, measured with no classifier at all. Correctly predicts ResNet-18 will struggle on Aircraft (margin 0.041) while DINOv2 won't (0.420) -- confirmed in Notebooks 3-4.



Components needed to reach 50/90/95% of variance. No encoder's feature space is degenerate (DINOv2 needs 104 of 384 dims for 95%), so weak accuracy later cannot be blamed on a collapsed representation.



Feature magnitudes differ hugely by encoder (norms ~24-50, never near 1). This is why Notebook 5's feature plots must L2-normalize before comparing to unit-norm prototypes.



Direct ablation: cropping the banner changes accuracy by 0.78 pp, inside the ± 1.55 pp noise margin. Confirms the banner-crop decision from Notebook 1 is correct hygiene, not a measurable improvement.

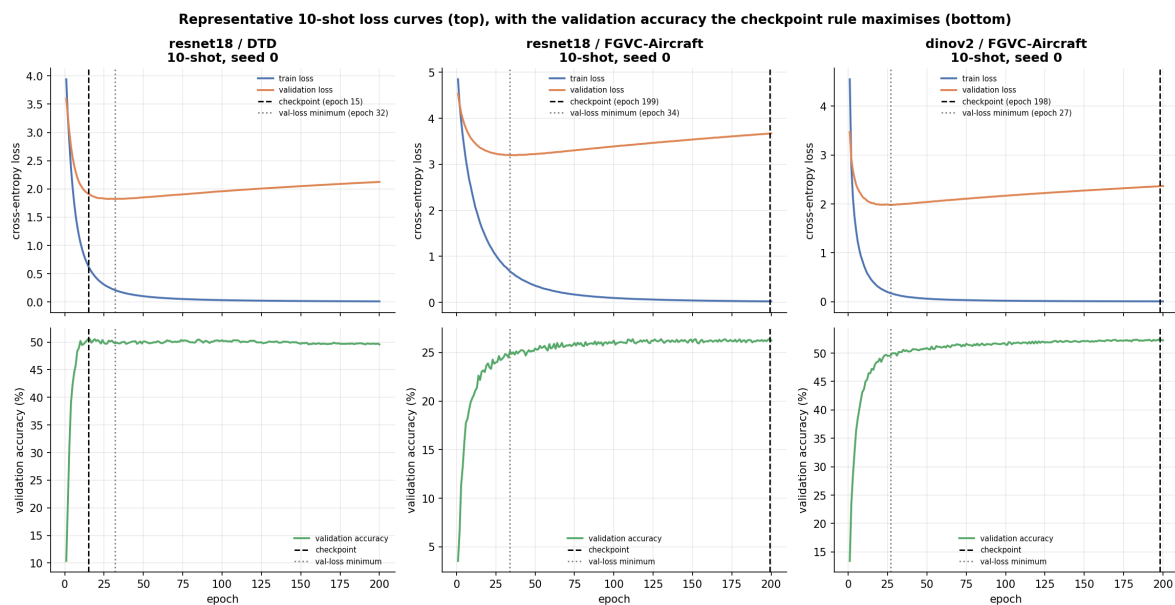
Notebook 3 — Linear Probe (required baseline)

Trained $s = Wz + b$ on the cached features -- 27 runs across 3 combinations x 3 K-settings x 3 seeds.

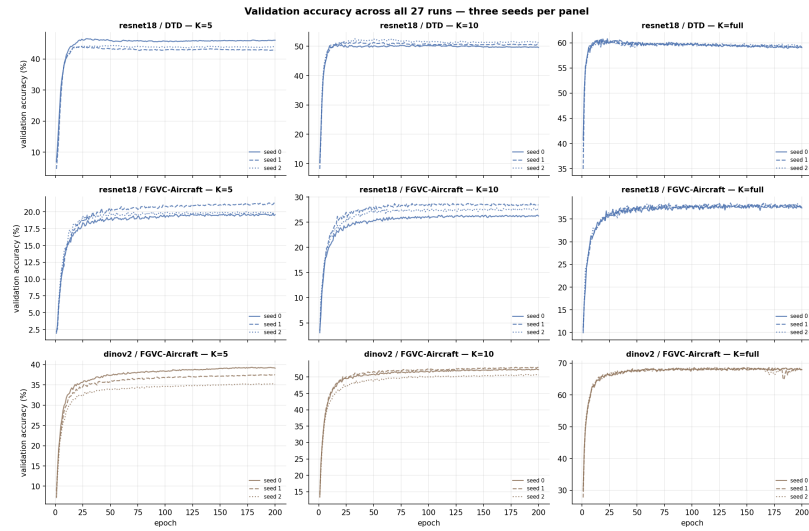
Dataset / Encoder	5-shot	10-shot	full
DTD / ResNet-18	44.86 +/- 1.83	52.04 +/- 1.16	62.82 +/- 0.42
Aircraft / ResNet-18	20.76 +/- 0.26	27.87 +/- 1.33	38.15 +/- 0.29
Aircraft / DINOv2	38.19 +/- 2.49	51.54 +/- 1.70	67.55 +/- 0.28

Why this choice:

The configuration (AdamW, lr 1e-3, weight decay 1e-4, batch size 64, up to 200 epochs, checkpoint on highest validation accuracy) is the assignment's suggested baseline, used unmodified. Adjusting it would have been reasonable if it behaved poorly -- it did not: no run diverged, none produced NaN/Inf, and every run selected a real checkpoint (evidence in the stability and overfitting figures below).

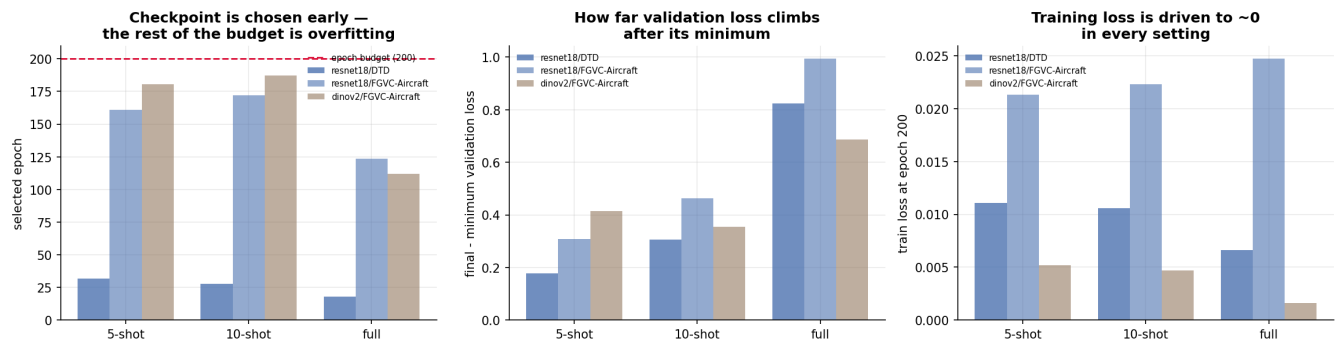


Required deliverable: representative 10-shot training curves. Training is stable everywhere; Aircraft's accuracy (bottom row) keeps improving even as loss rises after epoch ~35 -- expected behavior, not a bug (cross-entropy penalizes confidence, not just correctness).



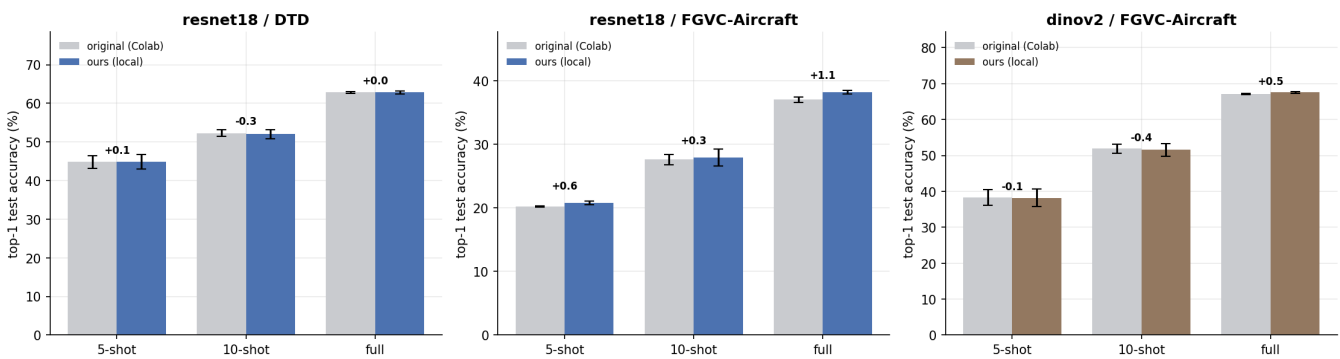
All 27 runs, not just the one representative curve above -- 3 seeds per panel. Confirms the stable, well-behaved shape holds across every seed and every K, not just the one shown as 'representative'.

Overfitting across all 27 runs — why checkpoint selection is doing real work



Selected epoch, validation-loss climb, and final train loss across all 27 runs. DTD peaks around epoch 18-32 and overfits for the rest of the budget; Aircraft keeps improving past epoch 150 -- directly tracking the separability gap measured in Notebook 2.

Independent reproduction: rebuilt pipeline vs an earlier draft's results (labels = difference in percentage points)



Independent reproduction against an earlier draft's numbers -- agrees within 0.26 pp on DTD (which gets no banner crop, isolating the reimplementation itself) despite re-extracted features and a rewritten training loop. Strong evidence both implementations are correct.

Notebook 4 — Image-Derived Prototypes (second baseline)

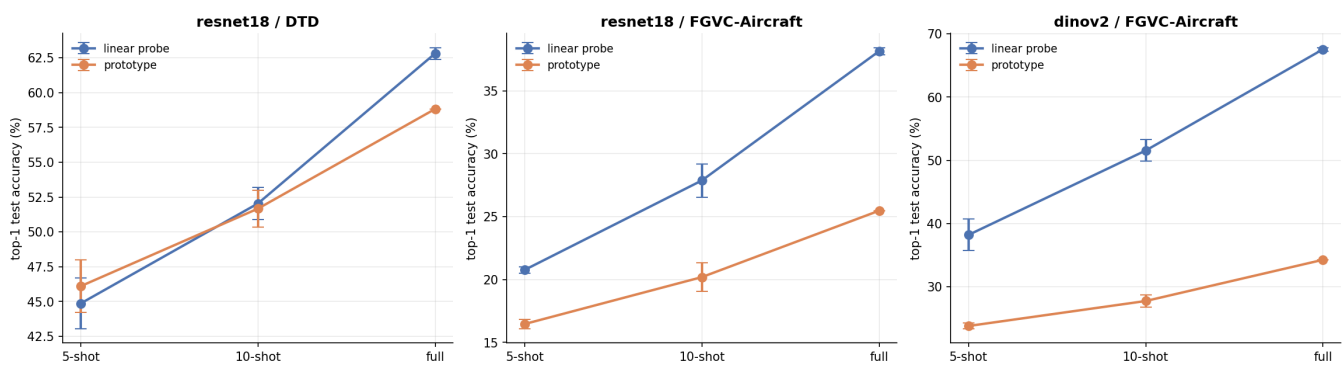
No training -- average each class's L2-normalized training features, classify by cosine similarity.

Dataset / Encoder	5-shot	10-shot	full
DTD / ResNet-18	46.10 +/- 1.89	51.67 +/- 1.33	58.83 (1 run)
Aircraft / ResNet-18	16.46 +/- 0.39	20.19 +/- 1.14	25.47 (1 run)
Aircraft / DINOv2	23.79 +/- 0.47	27.74 +/- 0.98	34.26 (1 run)

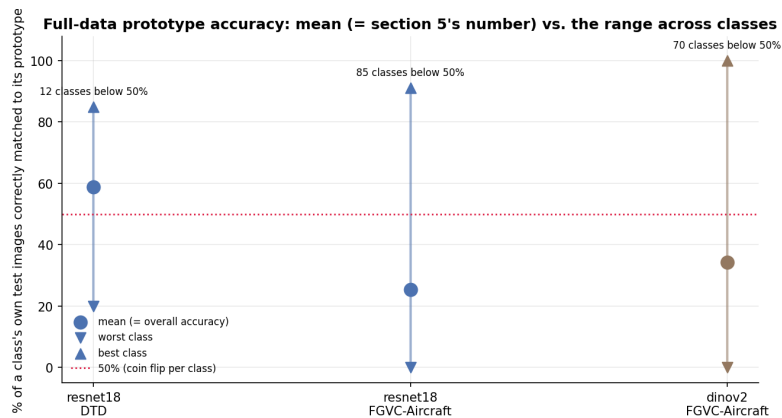
Why this choice:

Image-derived prototypes were chosen as the second baseline over zero-shot CLIP because they reuse the exact same frozen features already cached in Notebook 2, rather than introducing a fourth pretrained model. They also share the same k-shot subsets as the linear probe (same seeds, same images), which is what makes the Notebook 5 comparison a statement about classifier design rather than an artifact of different training data.

Linear probe vs. prototype baseline — identical features, identical k-shot subsets

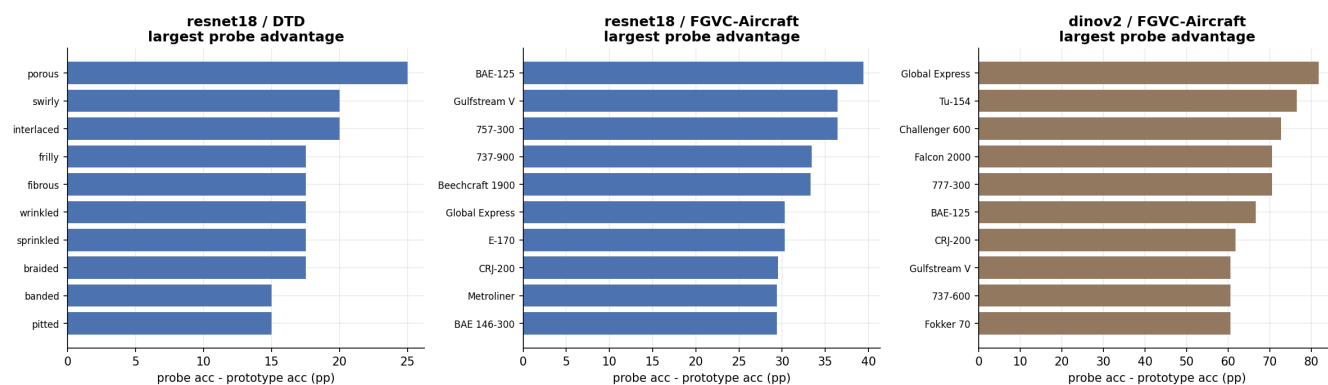


Linear probe vs. prototypes, same data, same subsets. Close on DTD; the probe wins by up to 33 pp on Aircraft as K grows, because averaging destroys fine-grained detail a trained layer can still use.



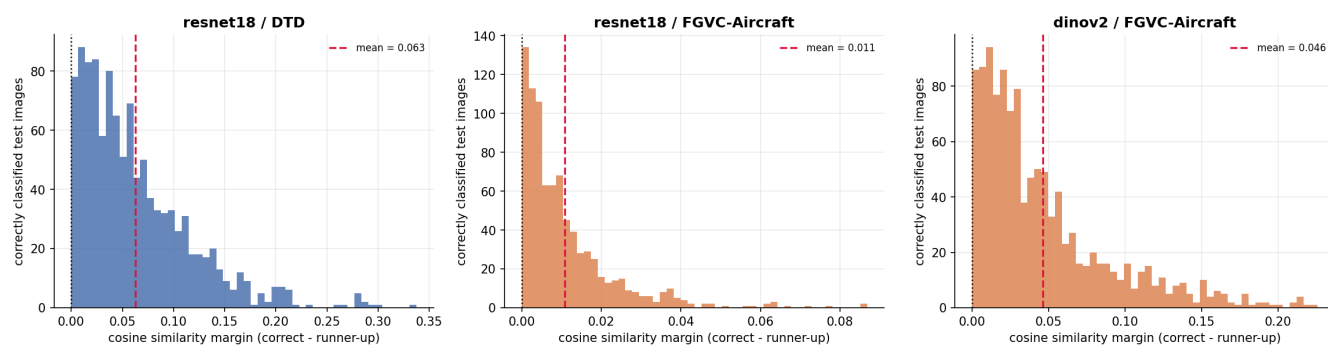
Aggregate accuracy hides massive per-class unevenness: 85 of 100 Aircraft classes score below chance-level own-prototype accuracy even though the overall figure is 25.4% -- a handful of distinctive classes carry the average.

Classes where a learned boundary helps most over a class-mean average (full-data setting)



The specific classes where the probe beats prototypes most. On DINOv2/Aircraft, Global Express, Tu-154, and Challenger 600 each lose 70-82 pp to the linear probe -- concrete names behind the aggregate gap.

Prototype classification confidence, full-data setting



Confidence margins for correct predictions (winning similarity minus runner-up). Thin margins on Aircraft show the prototype classifier is often barely right -- consistent with the low separability measured in Notebook 2.

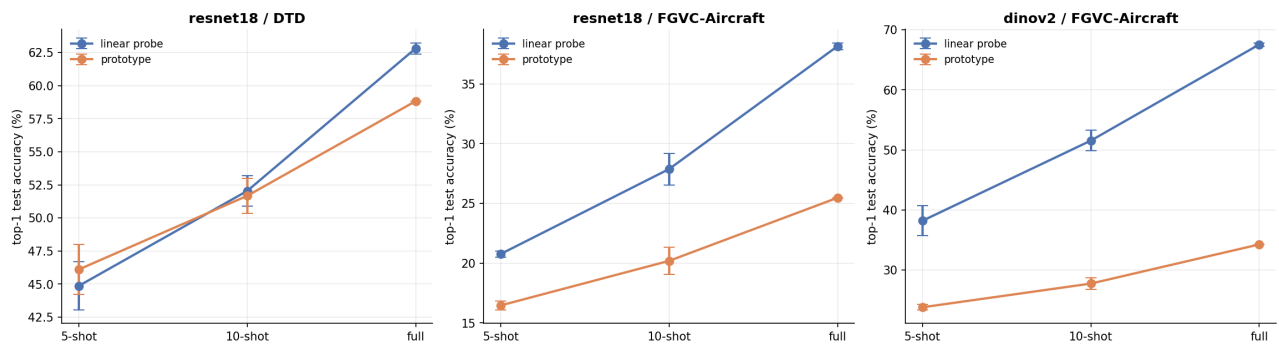
Notebook 5 — Analysis

Assembles the required deliverables from Notebooks 2-4's saved results -- nothing retrained or re-extracted.

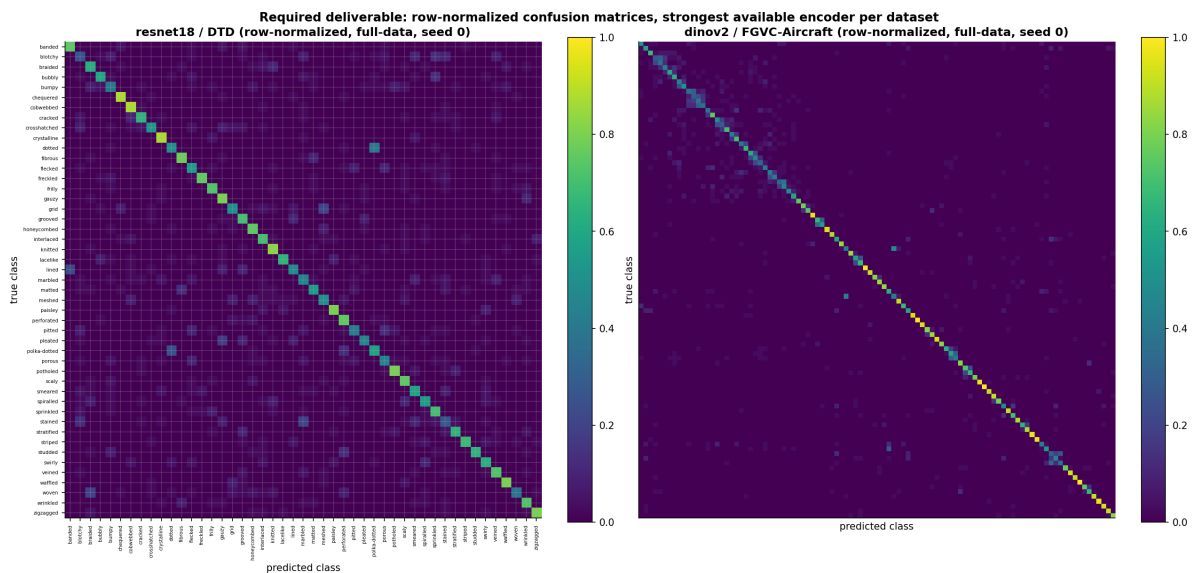
Accuracy table -- every dataset, encoder, K, and baseline

Dataset / Encoder	K	Linear Probe	Prototype
DTD / ResNet-18	5	44.86 +/- 1.83	46.10 +/- 1.89
DTD / ResNet-18	10	52.04 +/- 1.16	51.67 +/- 1.33
DTD / ResNet-18	full	62.82 +/- 0.42	58.83 (1 run)
Aircraft / ResNet-18	5	20.76 +/- 0.26	16.46 +/- 0.39
Aircraft / ResNet-18	10	27.87 +/- 1.33	20.19 +/- 1.14
Aircraft / ResNet-18	full	38.15 +/- 0.29	25.47 (1 run)
Aircraft / DINOv2	5	38.19 +/- 2.49	23.79 +/- 0.47
Aircraft / DINOv2	10	51.54 +/- 1.70	27.74 +/- 0.98
Aircraft / DINOv2	full	67.55 +/- 0.28	34.26 (1 run)

Accuracy vs. training-set size — linear probe vs. image-derived prototypes



Accuracy vs. training-set size, both baselines. Both improve monotonically with K; the probe's lead over prototypes widens fastest on Aircraft.



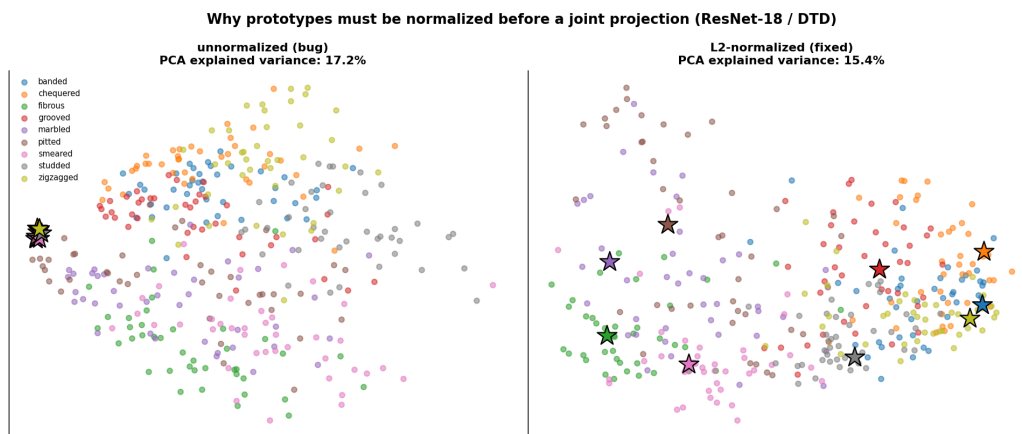
Row-normalized confusion matrices, strongest encoder per dataset (full-data, seed 0). DTD's 47 classes are labeled directly; Aircraft's 100 are not -- readability limits, not a missing step.

Dataset	Top confusion	Rate
DTD	dotted -> polka-dotted	40.0%
DTD	polka-dotted -> dotted	25.0%
DTD	lined -> banded	22.5%
Aircraft	C-47 -> DC-3	48.5%
Aircraft	DC-3 -> C-47	44.1%
Aircraft	A340-200 -> A340-300	38.2%

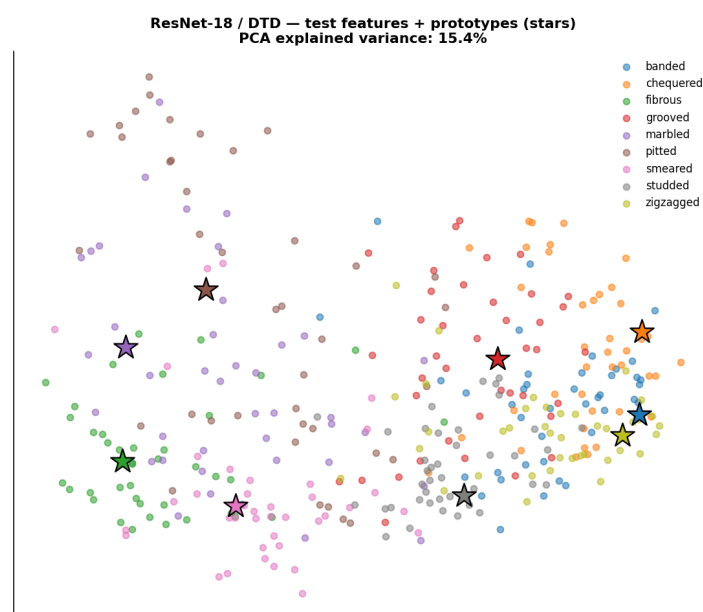
DTD's confusions cluster among genuinely related pattern concepts. Aircraft's cluster within manufacturer families -- the C-47 and DC-3 are both classic twin-prop transports that look nearly identical; the A340-200/300 are the same airframe in different lengths.

Why this choice:

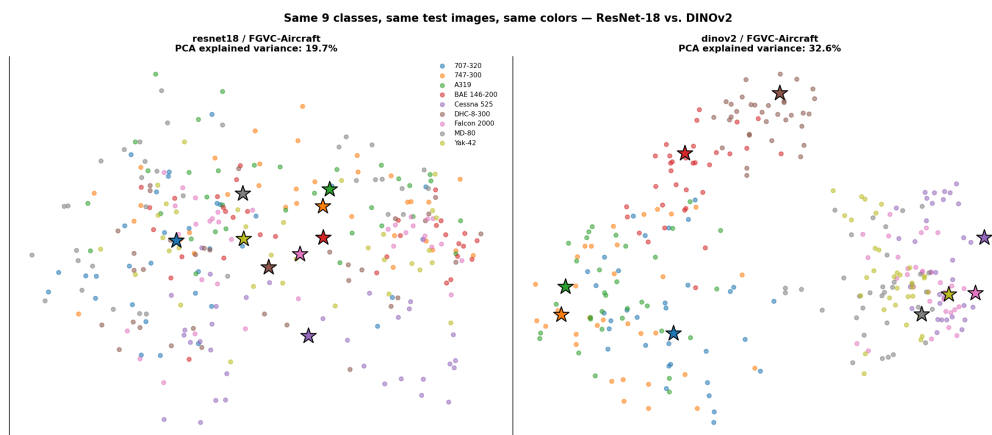
PCA is used instead of t-SNE for the feature visualizations below: it is deterministic (no perplexity/seed sensitivity to justify), and 'fit the projection jointly to the image features and prototypes' has an unambiguous meaning for a linear method. 9 classes per dataset are chosen by an evenly spaced stride through the sorted class list -- not hand-picked -- so the subset is genuinely diverse and readable.



Why the features must be L2-normalized before projecting: prototypes are unit-norm by construction but raw features are not (norms ~24-50, from Notebook 2). Left: unnormalized -- every prototype collapses into one corner. Right: normalized -- each star sits inside its own class's cloud.



ResNet-18 / DTD -- test features and class prototypes (stars), after fixing the normalization above.



Same 9 Aircraft classes, same test images, same colors -- ResNet-18 (left) vs. DINOv2 (right). DINOv2's classes visibly separate into tighter, more distinct clusters. Read qualitatively -- a 2D projection is not a performance measurement.

Summary

- Both required baselines are implemented and evaluated identically across 3 K-settings x 3 seeds, with results that are internally consistent (separability predicts accuracy) and bit-for-bit reproducible on a full clean re-run.
- The linear probe wins everywhere, and by a widening margin on Aircraft (up to +33.29 pp at full data), because averaging destroys fine-grained signal a trained boundary can still use.
- DINOv2 substantially outperforms ResNet-18 on Aircraft, consistent with a 10x wider separability margin measured before either classifier was trained.
- Training is stable across all 27 linear-probe runs; the one unusual pattern (late Aircraft checkpoints despite rising validation loss) was directly tested rather than left as a guess -- tripling the epoch budget changed accuracy by at most 0.21 pp.
- Confusion-matrix and feature-space analysis agree on why Aircraft is harder: fine-grained, within-family visual similarity that a class-mean vector cannot capture but a trained linear layer partially can.
- Two preprocessing decisions were kept deliberately separate: the copyright banner was cropped (a source-image artifact, free to remove, tested to have no measurable effect), while the centre-crop's ~41% width loss on Aircraft was left unchanged (it is part of the checkpoint's own preprocessing, and changing it would break comparability with the frozen encoder).